

3.3V/2A
Fsw=1.5MHz
L=1uH
Ic(RMS1) = 0.66A

$D = V_{out}/V_{in} = 3.3/5 = 0.66$
 $I_{Ripple} = (V_{out}(1 - V_{out}/V_{in})) / (L * f)$
 $I_{In} = I_{out} * (V_{out}/V_{in})$
 $I_{c(RMS)} = I_{out} * \sqrt{D * (V_{out}/V_{in})(1 - (V_{out}/V_{in}))}$
 $I_{c(RMS)tot} = \sqrt{I_{c(RMS1)}^2 + I_{c(RMS2)}^2}$

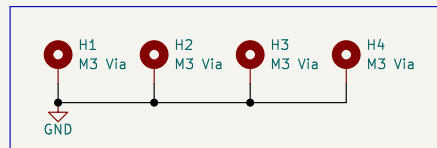
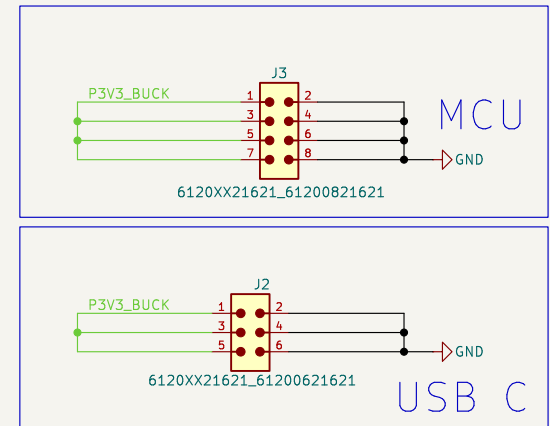
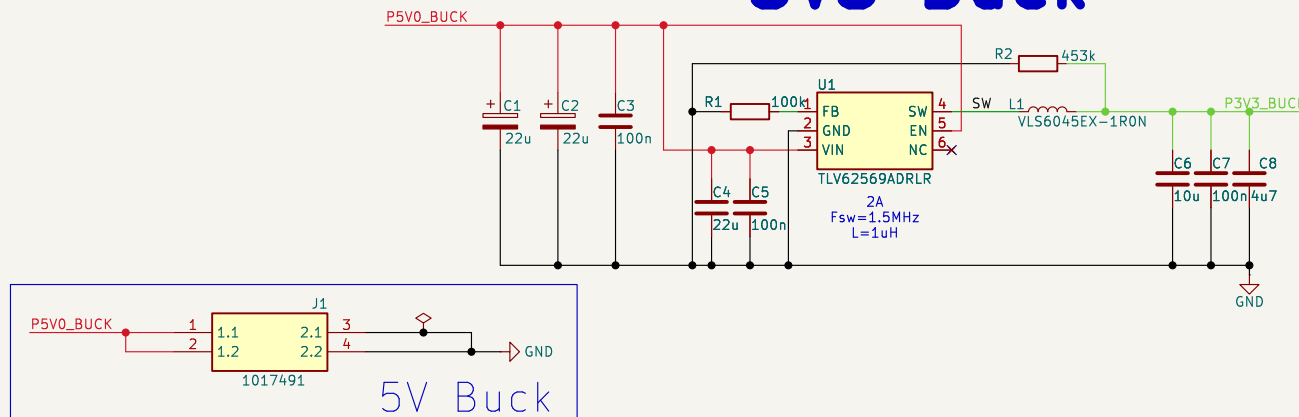
$I_{Ripple} = 0.748A$
 $I_{In} = 1.32A$
 $I_{c(RMS2)} = 0.947A$
 $I_{c(RMS)tot} = 1.155A \rightarrow 1.4A$

1, find cap bank > 1.4A rms ripple current

2, $Ripple_V_{out} = I_{Ripple} * ESR$
 $33mA = 0.748A * ESR, ESR=44.1mOhm$

3, 1.32A is drawn from the previous 5V stage

3V3 Buck



Supplies 3v3 volts, takes in 5 volts
Locked Inc.

Sheet: /
File: STAR_P3V3_BUCK.kicad_sch

Title: 3V3 Buck

Size: A4
KiCad E.D.A. 9.0.5

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